

T2

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Yholiño Walder Comun Perez

Y Holinto Comua Pronez

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad C = [1 \quad 2] \rightarrow \text{Afirmado}$$

(2) Matriz de Controlabilidad

$$C = [B \quad AB]$$

$$AB = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ -6 \end{bmatrix} ; \quad C = \begin{bmatrix} 2 & -2 \\ 3 & -6 \end{bmatrix}$$

$$\det(C) = (2)(-6) - (2)(3) = -12 + 6 = -6 \neq 0$$

$\text{Rango}(C) = 2 = n \rightarrow$ Sistema controlable

(3) Observabilidad: Afirmado $C = [1 \quad 2]$

$$O = \begin{bmatrix} C \\ CA \end{bmatrix}$$

$$CA = [C] \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} = [-2]$$

$$O = \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$$

$$\det(O) = (1)(-1) - (2)(-2) = -1 + 4 = 3 \neq 0$$

$\text{Rango}(O) = 2 = n \rightarrow$ Sistema observable

```

Co =
    2    -2
    3    -6

rango(Co) = 2
det(Co)    = -6.0000
Controlable

Ob =
    1    1
   -1   -2

rango(Ob) = 2
det(Ob)    = -1.0000
Observable

```

%% Controlabilidad y Observabilidad - Sistema Discreto
 clc; clear; close all;

%% Sistema

```

A = [-1  0;
     0 -2];

```

```

B = [2; 3];

```

```

C = [1  1];

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D = 0;

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n = size(A,1);

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%% Controlabilidad

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Co = ctrb(A,B);

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disp('Co =');

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disp(Co);

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fprintf('rango(Co) = %d\n', rank(Co));

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fprintf('det(Co) = %.4f\n', det(Co));

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if rank(Co) == n

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    fprintf('Controlable\n\n');

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else
    fprintf('No controlable\n\n');
end

%% Observabilidad
Ob = obsv(A,C);

disp('Ob =');
disp(Ob);
fprintf('rango(Ob) = %d\n', rank(Ob));
fprintf('det(Ob) = %.4f\n', det(Ob));
if rank(Ob) == n
    fprintf('Observable\n\n');
else
    fprintf('No observable\n\n');
end
```